Introduction to Robotics

What is a Robot?

What do you think a robot *actually* is?

Where have you seen robots?

Are all of these robots?



RC Car



CNC Machine



Google Assistant / Siri



Self-Driving Car

Definition of a Robot

International Organization for Standardization (ISO) Definition (ISO 8373:2012):
"An actuated mechanism programmable in two or more axes with a degree of autonomy, moving within its environment, to perform intended tasks."

Merriam-Webster Dictionary:

a machine that resembles a living creature in being capable of moving independently (as by walking or rolling on wheels) and performing complex actions (such as grasping and moving objects)

A robot is a machine—especially one programmable by a computer—capable of carrying out a complex series of actions automatically (**Wikipedia, Oxford Dictionary**)

Definition of a Modern-Day Robot

A modern robot is an intelligent electromechanical system that perceives its environment, processes information, and executes physical actions with varying degrees of autonomy.

Every robot must have three core capabilities:

1. **Sense:** Collect information using sensors(e.g., cameras, LiDAR, encoders, force sensors)
2. **Think:** Process information or make decisions(e.g., controller, microprocessor, AI logic)
3. **Act:** Perform physical actions(e.g., motors, wheels, joints, end-effectors)

sense–think–act

If a machine **cannot sense, think, and act,** then it is **not considered a robot.**

Are all of these robots?



RC Car

Remote-Controlled: *Act but do not sense or process information.*



CNC Machine

Automated: *Act but do not make decisions based on the environment.*
Has sensors for precision, but follows fixed instructions and does not adapt to environment.



Google Assistant / Siri

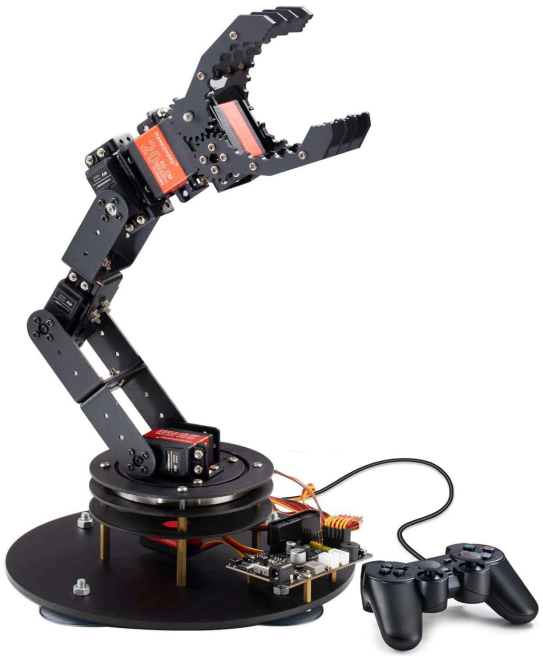
Intelligent: *Process information, but no physical actions.*



Self-Driving Car

Fully Autonomous: Senses environment, makes decisions and acts

Is a manipulator with NO sensors still a robot?



"A robot is a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks."
— Robot Institute of America (RIA)

Yes. According to classical industrial robotics definitions (including RIA), a manipulator **does not need sensors** to be considered a robot.

Not fully. Modern robotics emphasizes **autonomy.**

Robot = Sense + Plan + Act

A manipulator without sensors is a **classical robot**, but **not** an **autonomous or intelligent robot**.

Robot, Robotics, Robotics Engineering

Robot

A programmable machine that **senses, thinks, and acts** in the physical world with some autonomy.

Robotics

The science and technology of **designing, controlling, and understanding robots**.

Robotics Engineering

The engineering discipline that **creates functioning robotic systems** by integrating mechanics, electronics, computing, and intelligence

Robotics brings together multiple fields: Mechanical design, Electronics and sensors, Computing and AI, Control systems, Perception and machine learning, Human–robot interaction

Answers questions like:

- How do robots move?
- How do they sense the world?
- How do they plan motion?
- How do they learn and adapt?
- How do they interact safely with humans?

Robotics Engineering applying principles from: Mechanical engineering, Electrical/electronic engineering Computer engineering, Control and automation, Artificial intelligence…

A **robotics engineer:**

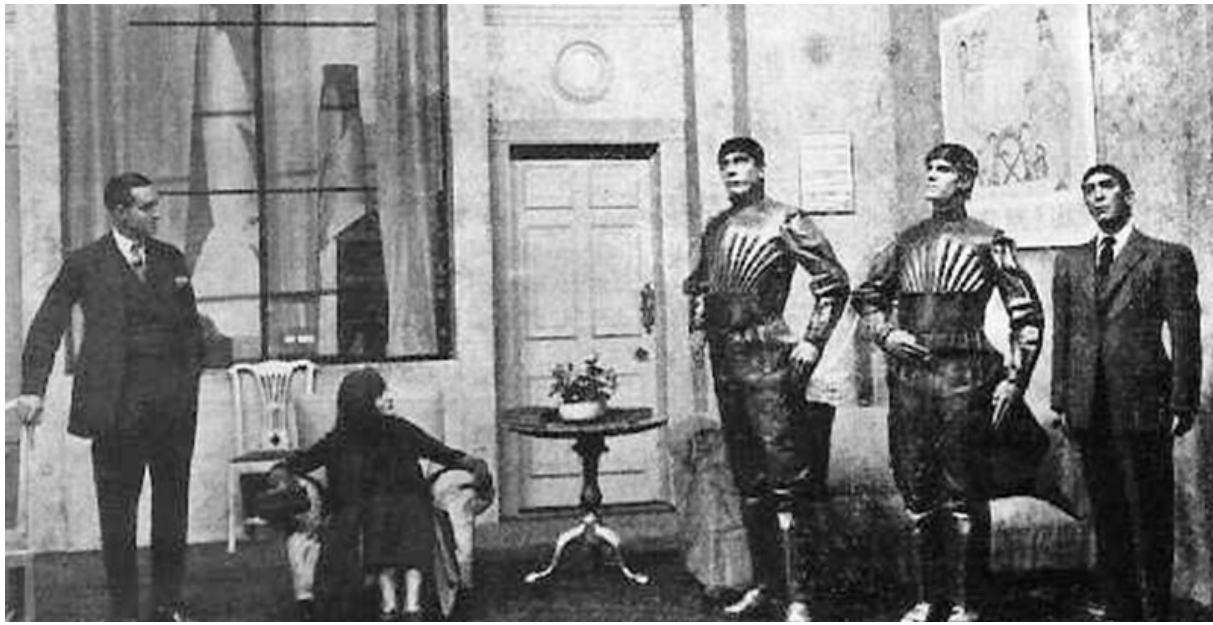
- Designs robot mechanisms
- Develops hardware and electronics
- Integrates sensors and actuators
- Programs control algorithms
- Builds autonomous behavior
- Ensures safety and performance
- Optimizes robots for real-world applications

A Brief History of Robotics

1920 – The Word "Robot"

- Introduced by a Czech writer **Karel Čapek** in a science fiction Play *R.U.R. – Rossum's Universal Robots (English Translation)*

- Origin: **"robota" = forced labor**
- First time the idea of mechanical workers entered mainstream culture



A scene from the play, showing three robots
(PC: https://en.wikipedia.org/wiki/R.U.R).

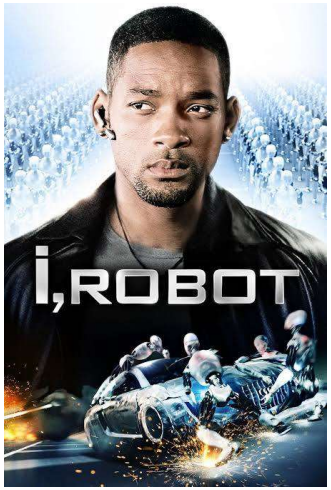
A Brief History of Robotics

1940s – Isaac Asimov & Robot Ethics

- Isaac Asimov coined the term **'robotics'** in his 1941 short story 'Liar!', later included in his book **I, Robot**.



Isaac Asimov

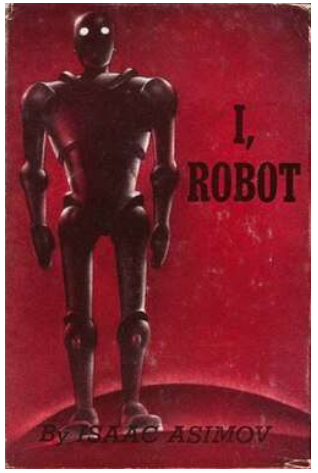


2004 movie based on I, Robot (1950) fiction by Asimov

- Introduced the **Three Laws of Robotics**
- Influenced modern thinking about robot behavior, safety, and ethics

Isaac Asimov's **Three Laws of Robotics**

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm (*Human safety*)
2. A robot must obey the orders given it by human beings, except where such orders would conflict with the First Law. (*Robots are slaves*)
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law. (*Survival of robots*)



A Brief History of Robotics

1954 – First Industrial Robot

- **George Devol** invents **Unimate**, the first programmable robot
- Later used in automotive manufacturing (GM, 1961)
- Start of modern robot manipulators



George Charles Devol Jr. (February 20, 1912 – August 11, 2011) was an American inventor, best known for creating **Unimate**, the first industrial robot. Devol's invention earned him the title "Grandfather of Robotics".

1970s–1990s – Industrial & Mobile Robotics

- Emergence of SCARA, Delta, and 6-DOF robot arms
- Automated welding, painting, assembly
- Early mobile robots and research platforms (Stanford Cart, Shakey)
- **Shakey** was the first general-purpose mobile intelligent robot developed by Stanford Research Institute (SRI).



Shakey was developed from approximately 1966 through 1972 with Charles Rosen, Nils Nilsson and Peter Hart as project managers

Modern Day Robots

2000s: -- Service & Medical Robc

- Robotic surgery (da Vinci)
- Warehouse automation
- Humanoid research (Honda ASIMO)









ASIMO (Advanced Step in Innovative Mobility) is a humanoid robot created by Honda in 2000.

2010s–Now: AI, Cobots & Autonomous Systems

- Drones, AGVs, AMRs, Humanoid Robots
- Integration of AI, vision, reinforcement learning
- Robots in homes, hospitals, agriculture & logistics





Nurse Assistant Robot "Nurabot“ (2025)









Current Robotics Research: **Physical AI (Embodied AI)**

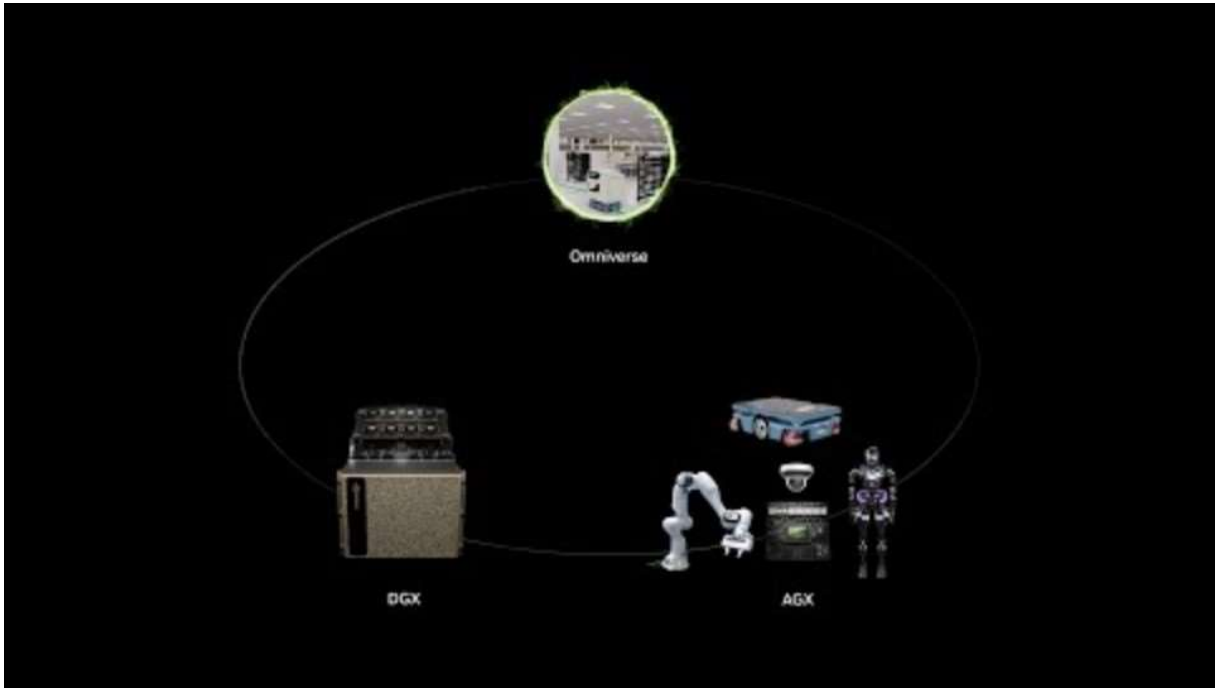
Physical AI or Embodied AI = AI that doesn't just think, but it acts in the real world.

Key Components of Physical AI

- **Embodiment:** The AI is inside a physical body (robot arm, drone, mobile robot, humanoid).
- **Perception:** Robots use sensors, cameras, and AI models to interpret the world.
- **Learning & Adaptation:** Robots improve from experience (reinforcement learning, imitation learning).
- **Control & Action:** AI generates motion, manipulation, and decision-making in real time.
- **Human–Robot Interaction:** Robots understand and respond to humans safely and intelligently.

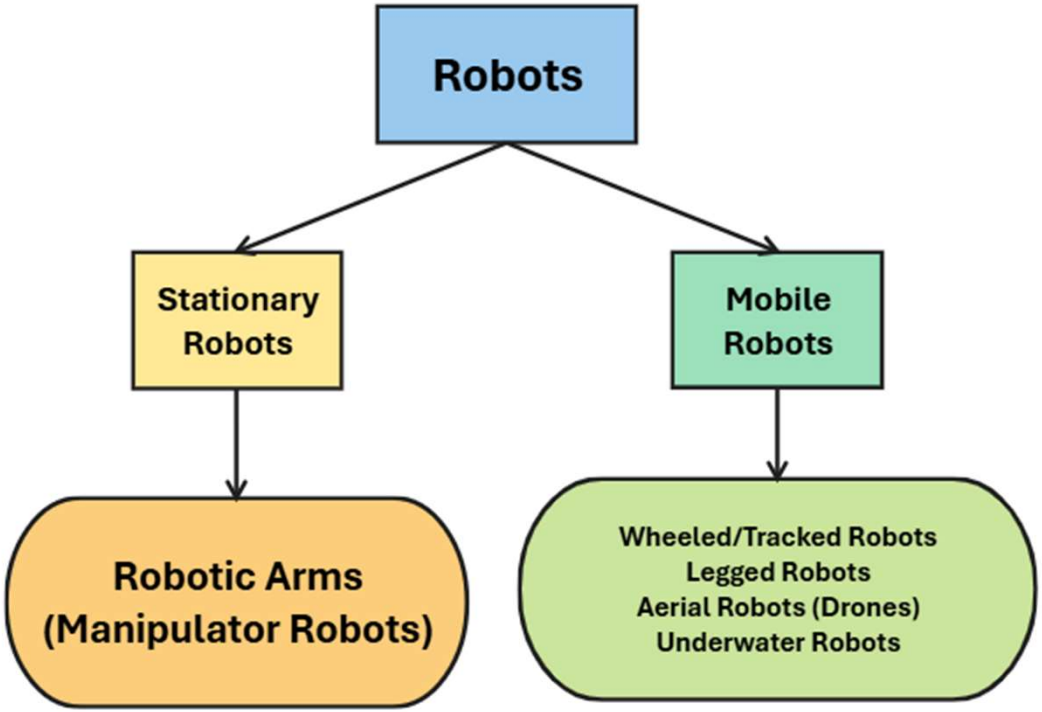


The Next Wave of AI: Physical AI” by NVIDIA



Watch Video

Classification of Robots based on the mobility



Terminologies in Robotics

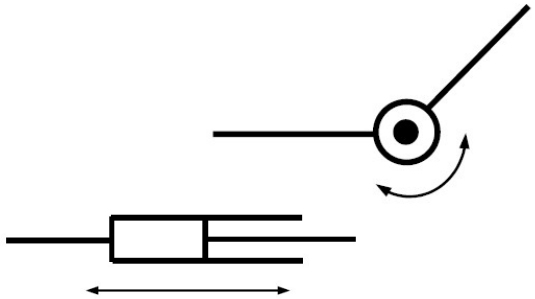


Manipulator (Robot Arm): A robot made of links and joints designed to move objects or perform specific tasks.

Joint: The part of a robot that moves.

Types:

- **Revolute** (rotation)
- **Prismatic** (linear motion)

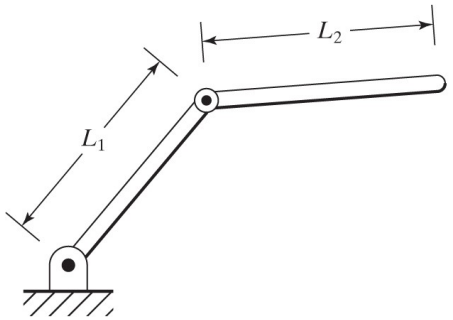


Link

A rigid segment between two joints. A robot is a chain of links connected by joints.

End Effector: The tool attached to the robot's arm, such as a gripper, welder, or sensor, that actually interacts with objects.





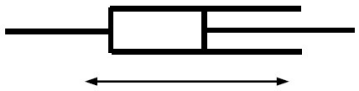
Two-link manipulator with link lengths L1 and L2.; and two revolute joints.

Degrees of Freedom (DoF)

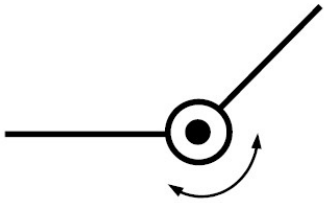
Degrees of Freedom (DOF): The number of independent movements or axes of rotation a robot can make.

Each DOF corresponds to one of the robot's:

- **Translations**: Move along X, Y, or Z
- **Rotations**: Rotate about X, Y, or Z



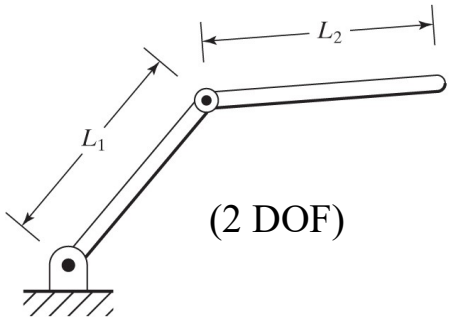
Prismatic joint (1 DOF)



Rotational joint (1 DOF)

Why DOF Matters?

- Determines robot's **flexibility**
- Defines how many coordinates are needed to describe its pose
- Impacts control, kinematics, and ability to reach tasks

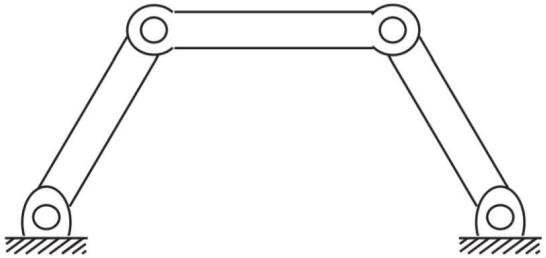


(2 DOF)

Examples

- A simple hinge joint: **1 DOF** (rotation)
- A planar 2-link arm : **2 DOF**
- A typical industrial arm: **6 DOF** (3 for position, 3 for orientation)

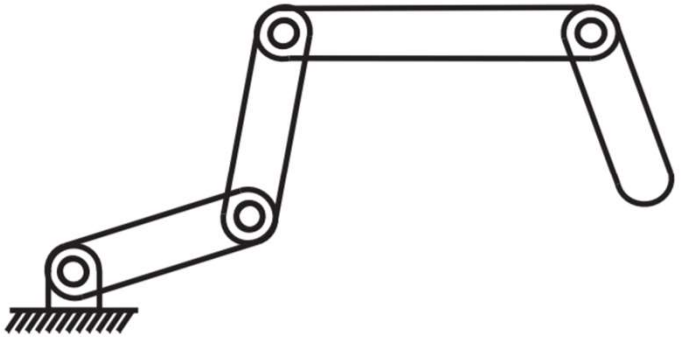
DOF -Examples



4-bar mechanism

Fixed connections to ground limits motion

- Due to the closed-chain geometry of the mechanism, motion initiated at any single revolute joint uniquely determines the configuration of the remaining joints.
- The joint variables are not independently controllable.
- The mechanism possesses **only one independent mode** of motion, yielding a total of **1 DOF**.



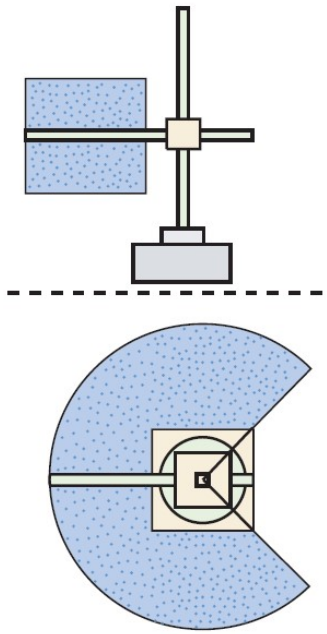
4-link planar serial chain

The mechanism forms an open-chain serial linkage consisting of four revolute joints connected end-to-end.
Because **each revolute joint contributes one independent rotational degree of freedom,** This is a 4-DOF serial manipulator.

Robot Workspace

The **total volume or area** that a robot's end-effector can reach, given its joint limits and link lengths.

- Ensures the robot can reach required points
- Helps in layout planning (factory floor, lab setup)
- Influences task feasibility and safety zones

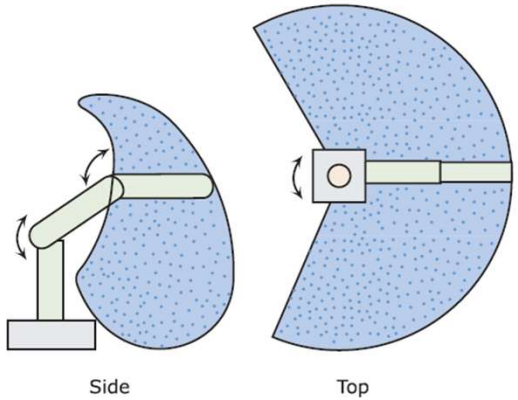


Cylindrical robot

Shapes of Workspaces is determined by the robot configuration

- Planar arms → circular shapes
- SCARA → donut-shaped (cylindrical)
- 6-DOF arms → spherical/cylindrical hybrids

Robot Workspace animation



Articulated robot

Frames and Coordinate Systems

A coordinate system is Reference system with X, Y, Z axes used to describe positions and orientations in space.

Frames associate these coordinate systems with specific points, such as joints or the robot's end-effector.

Frame types:

- Base frame (origin of robot)
- Joint frames
- End-effector frame
- World frame (lab or environment)

Right-hand rule (defines axis directions)

